Investigating the Effect of Natural Treatments Quercetin and Resveratrol on Cell Viability in BxPC-3 Pancreatic Cancer Cells

Cellular & Molecular Biology (CELL)

BACKGROUND

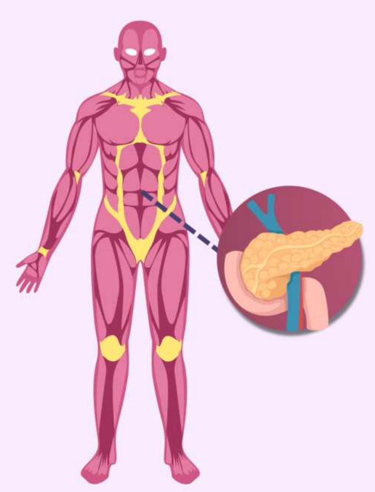


Image made by researcher on Canva

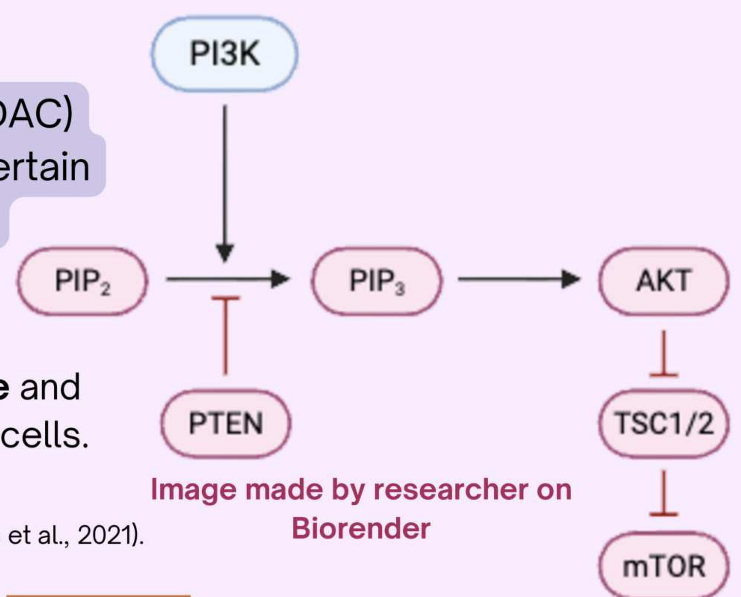
11th most commonly diagnosed cancer in the U.S.

Abnormal cells in the pancreas **uncontrollably grow** and divide.

Most are **exocrine**; **adenocarcinoma** is most common (highly **aggressive**, contains **complex** pathways, and is often **resistant** to standard pathways) (What Is Pancreatic Cancer?, n.d.).

Pancreatic ductal adenocarcinoma (PDAC) can hyperactivate certain oncogenic signaling pathways

Pathways promote therapeutic **resistance** and **proliferation** of tumor cells. An example would be PI3K/AKT/mTOR (Rascio et al., 2021).



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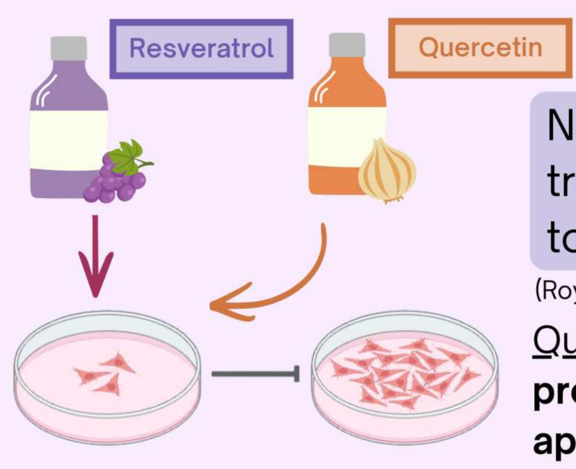

Natural compounds as treatments have reduced toxicity and proliferation (Roy et al., 2025).

Quercetin + Resveratrol **inhibit proliferation**/promote **apoptosis** (Tezerji et al., 2022).

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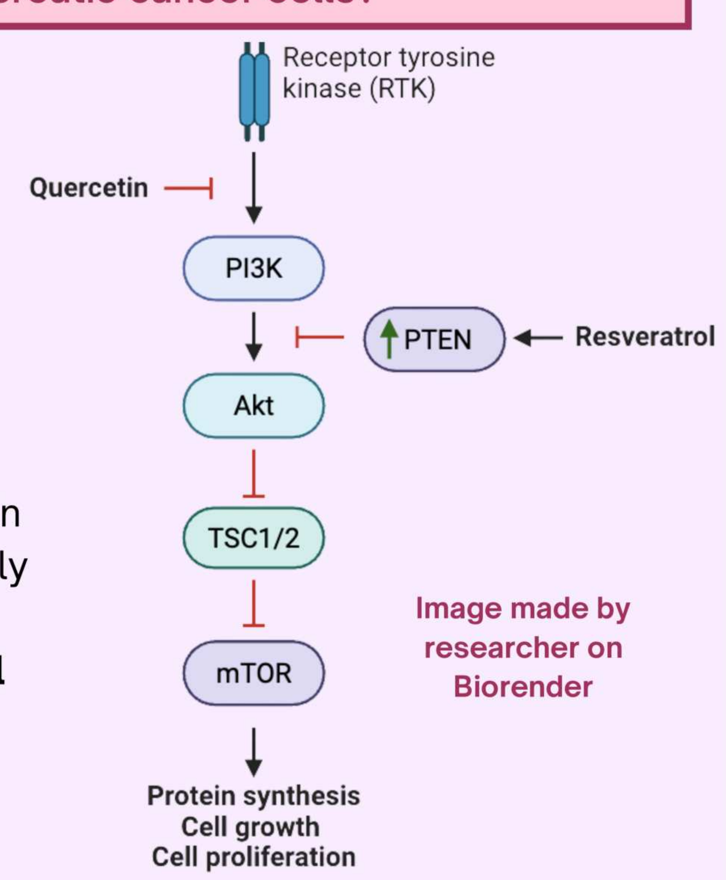
HYPOTHESIS

What are the optimal dosages of quercetin and resveratrol when used as a combination therapy at which cell viability is most significantly reduced in BxPC-3 pancreatic cancer cells?

Highest concentrations of resveratrol + quercetin (**30µM R + 40µM Q**)

Significant **reduction** in **cell viability,** potentially due to reduced **proliferation** and **cell death** from **higher concentrations** of compounds



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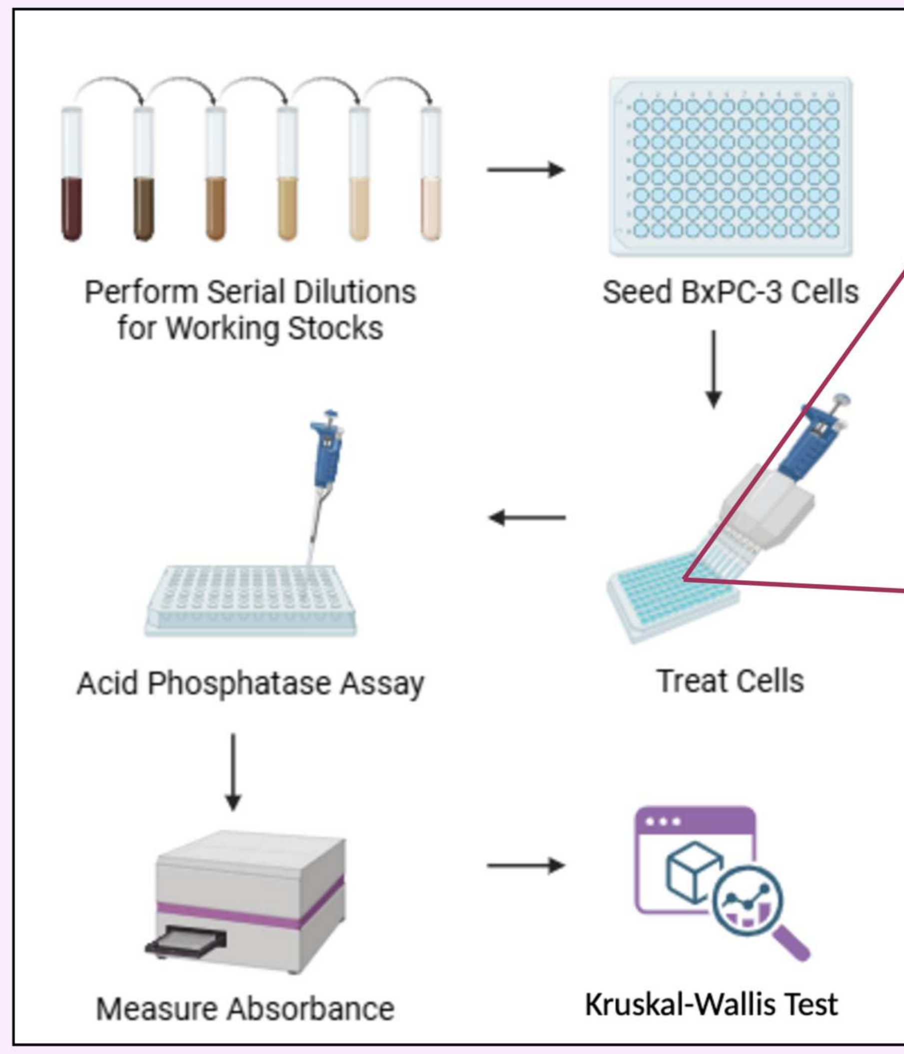
METHODOLOGY



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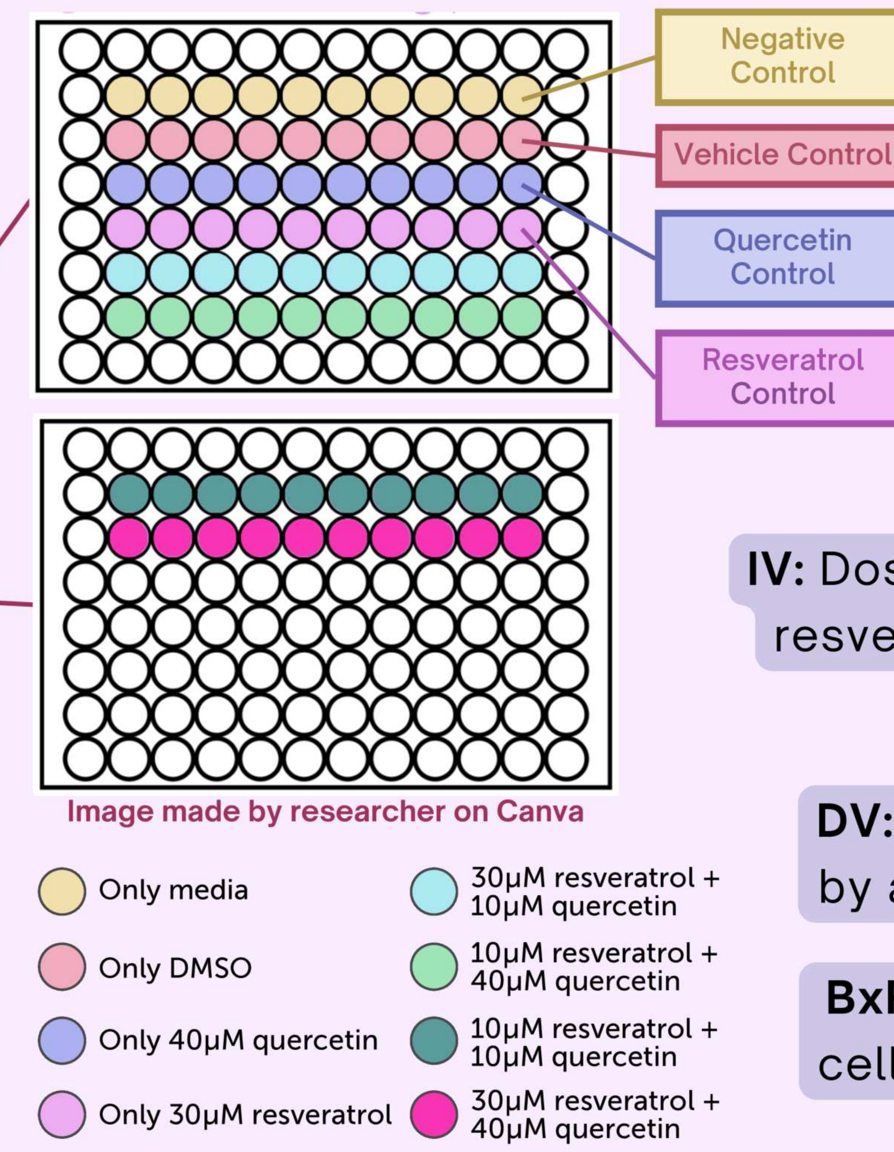

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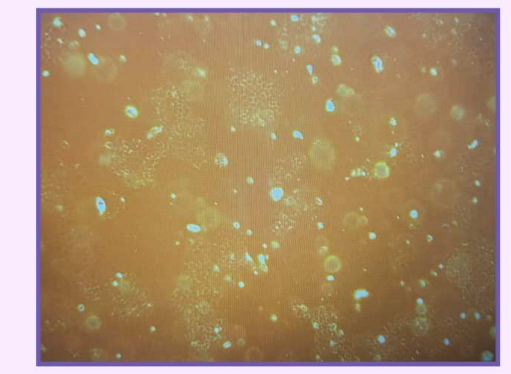


Image of BxPC-3 Cells (70% confluent, 400X magnification) taken by researcher

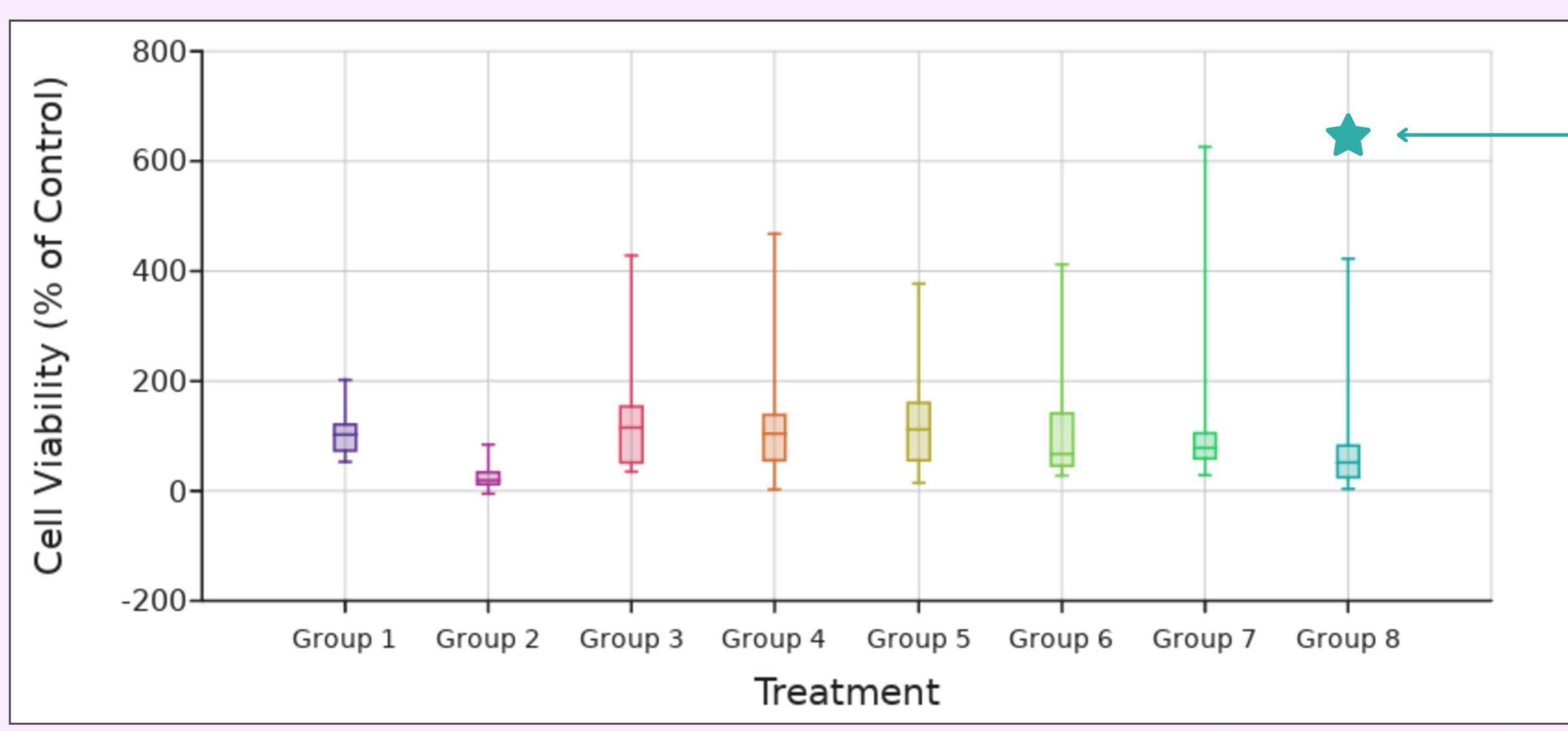
IV: Dosage levels of quercetin + resveratrol, prepared in stock solutions

DV: Cell viability; measured by acid phosphatase assay

BxPC-3 pancreatic cancer cells; human; adherent cell line

RESULTS

CELL VIABILITY



Star represents group hypothesized to have the lowest cell viability

Group 1 = Negative Control
Group 2 = Vehicle Control
Group 3 = 30µM R
Group 4 = 40µM Q
Group 5 = 30µM R + 10µM Q
Group 6 = 10µM R + 40µM Q
Group 7 = 10µM R + 10µM Q
Group 8 = 30µM R + 40µM Q

Figure 1: Cell viability was measured with an acid phosphatase assay. R stands for resveratrol, and Q stands for quercetin.

Image made by researcher on DataClassroom

Star represents group hypothesized to have the lowest cell viability

Image made by researcher on DataClassroom & Excel

KRUSKAL-WALLIS POST-HOC RESULTS

Comparison	P-Value	Result
All Treatments vs. Negative Control (N)		
30 µM R vs. N	1.00	Not significantly different
40 µM Q vs. N	1.00	Not significantly different
30µM R + 10µM Q vs. N	1.00	Not significantly different
10µM R + 40µM Q vs. N	1.00	Not significantly different
10µM R + 10µM Q vs. N	1.00	Not significantly different
30µM R + 40µM Q vs. N	0.02	Significantly different
30µM R + 40µM Q Comparisons With Significant or Potential Significant Difference		
30 µM R vs. 30µM R + 40µM Q	0.01	Significantly different
40 µM Q vs. 30µM R + 40µM Q	0.06	Some confidence of difference
30µM R + 10µM Q vs. 30µM R + 40µM Q	0.03	Significantly different

Table 1: Kruskal-Wallis test based on data with post-hoc results. Each group is compared to the negative control.

DISCUSSION

Cell Viability Results

- **30µM R + 40µM Q** = only group to **significantly reduce** cell viability compared to negative control (p = 0.02)

Collected Data **Supports** Hypothesis

- ★ = more effective than **resveratrol alone** (p = 0.01) and **30µM R + 10µM Q** (p = 0.03)
  - Suggests **quercetin** concentration **matters**
- **No other group** significantly reduced viability compared to negative control
- ● = **some confidence** of difference (p = 0.06), limiting conclusions about quercetin's **individual contribution**

Limitations

- **High variability** between replicates across trials likely **prevented** statistical significance
- Only **two** concentration levels per compound tested

Implications

- Could inform preclinical studies exploring **natural compound-based combination** therapies as alternative to chemotherapy
  - **Lower-toxicity** alternative if **optimized**
  - Can **minimize** adverse side effects

FUTURE WORK

- Observing specific mechanisms that occur as a result of the treatment
  - **Caspase-3/7 assay** to measure **apoptosis**
  - **Western blot** to measure specific **proteins** involved, such as mTOR (a protein kinase)

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